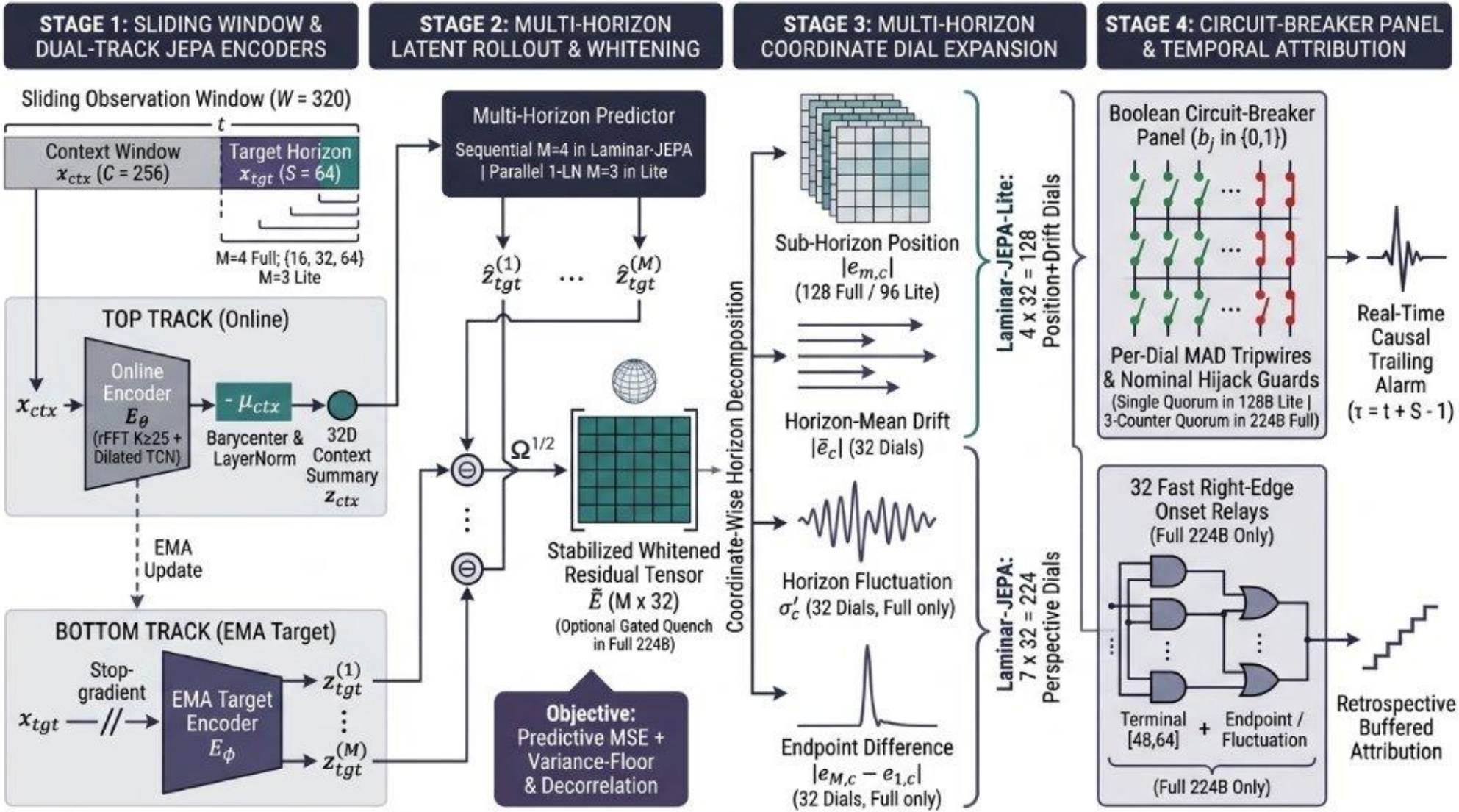

STAGE 1: SLIDING WINDOW & DUAL-TRACK JEPA ENCODERS
STAGE 2: MULTI-HORIZON LATENT ROLLOUT & WHITENING
STAGE 3: MULTI-HORIZON COORDINATE DIAL EXPANSION
STAGE 4: CIRCUIT-BREAKER PANEL & TEMPORAL ATTRIBUTION
Sliding Observation Window (W = 320)
t
Context Window x_{ctx} (C = 256)
Target Horizon x_{tgt} (S = 64)
M=4 Full; {16, 32, 64}
M=3 Lite
TOP TRACK (Online)
x_{ctx}
Online Encoder E_θ
(rFFT K≥25 + Dilated TCN)
$-\mu_{ctx}$
Barycenter & LayerNorm
32D Context Summary z_{ctx}
EMA Update
BOTTOM TRACK (EMA Target)
x_{tgt}
Stop-gradient
EMA Target Encoder E_ϕ
$z_{tgt}^{(1)}$
$z_{tgt}^{(M)}$
Multi-Horizon Predictor
Sequential M=4 in Laminar-JEPA
| Parallel 1-LN M=3 in Lite
$\hat{z}_{tgt}^{(1)}$ ... $\hat{z}_{tgt}^{(M)}$
$\Omega^{1/2}$
Stabilized Whitened Residual Tensor $\tilde{E}$ (M x 32)
(Optional Gated Quench in Full 224B)
Objective:
Predictive MSE + Variance-Floor & Decorrelation
Coordinate-Wise Horizon Decomposition
Sub-Horizon Position $|e_{m,c}|$
(128 Full / 96 Lite)
Horizon-Mean Drift $|\bar{e}_c|$ (32 Dials)
Horizon Fluctuation σ'_c (32 Dials, Full only)
Endpoint Difference $|e_{M,c} - e_{1,c}|$
(32 Dials, Full only)
Laminar-JEPA-Lite: 4 x 32 = 128 Position+Drift Dials
Laminar-JEPA: 7 x 32 = 224 Perspective Dials
Boolean Circuit-Breaker Panel (b_j in {0,1})
Per-Dial MAD Tripwires & Nominal Hijack Guards
(Single Quorum in 128B Lite | 3-Counter Quorum in 224B Full)
32 Fast Right-Edge Onset Relays
(Full 224B Only)
Terminal [48,64]
+
Endpoint / Fluctuation
(Full 224B Only)
Real-Time Causal Trailing Alarm
(τ = t + S - 1)
Retrospective Buffered Attribution